The Finite Ising/Jones B_3 Image
Lift-dependent and projectively intrinsic sigma–MGFs

Computational verification note

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Abstract

We test the finite-image formulas on the two-dimensional Ising/Jones braid representation. The displayed braid generators close to a 192-element unitary lift with scalar center of order eight. We verify its one-sided sigma–MGF and the full-twist selection rule. We then pass to conjugation on $\mathrm{End}(\mathbb{C}^2)$, obtaining the canonical 24-element projective image in dimension four, and verify the projective formula there.

1 The finite braid image

Put

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \qquad q = e^{-\pi i/8},$$

and define

$$\rho(\sigma_1) = qS, \qquad \rho(\sigma_2) = qHSH.$$

Direct multiplication gives the braid relation. Breadth-first matrix closure gives $|G| = 192$; quotienting its eight scalar matrices gives a projective image of order 24. The B_3 full twist is

$$\rho((\sigma_1\sigma_2)^3) = e^{-\pi i/4} I.$$

It therefore acts on $W_\tau = \bigotimes_a \mathrm{Sym}^{\tau_a}\mathbb{C}^2$ by $e^{-\pi i|\tau|/4}$, proving the necessary selection rule

$$\boxed{W_\tau^G = 0 \quad \text{unless} \quad 8 \mid |\tau|.}$$

The rule is necessary, not sufficient, as the degree-eight computations show.

2 Finite-image Molien proof

For every finite unitary image $G \leq U(V)$,

$$\mathcal{S}_{G,V}(\mathbf{t}) = \frac{1}{|G|}\sum_{g\in G}\prod_{a\geq 1}\det(I - t_a g)^{-1}.$$

Since the coefficient of t^r is the character of $\mathrm{Sym}^r V$, Reynolds averaging proves

$$[m_\tau]\mathcal{S}_{G,V} = \dim(W_\tau^G).$$

Taking a logarithm gives

$$\det(I - tg)^{-1} = \exp\left(\sum_{k\geq 1}\frac{t^k}{k}\mathrm{Tr}(g^k)\right),$$

and partitioning the sum into conjugacy classes proves the character/power-map and centralizer-weighted class formulas. No information about the infinite kernel of $B_3 \to G$ is needed after the finite image is fixed.

3 Projective correction

Changing a lift g to zg changes its eigenvalues, so the one-sided series is not a projective invariant. Conjugation on $\mathrm{End}(V)$ is unchanged:

$$\mathrm{Ad}_{zg}(A) = (zg)A(zg)^{-1} = gAg^{-1}.$$

Thus the projectively intrinsic series is the ordinary finite Molien series of the adjoint representation,

$$\mathcal{S}^{\mathrm{proj}}(\mathbf{t}) = \frac{1}{|\overline{G}|} \sum_{\bar{g}\in\overline{G}} \prod_a \det(I - t_a \,\mathrm{Ad}_{\bar{g}})^{-1}.$$

If g is any lift, then

$$\mathrm{Tr}(\mathrm{Ad}_g^k) = \mathrm{Tr}(g^k)\overline{\mathrm{Tr}(g^k)} = |\chi(g^k)|^2,$$

which proves the proposed projective power-trace formula and its independence of lift.

4 Independent matrix checks

For each τ, the program explicitly restricts tensor powers to normalized symmetric occupation bases and ranks the averaged Reynolds matrix. Separately, it computes $h_r(g)$ from traces of powers using Newton's recurrence and averages $\prod_a h_{\tau_a}(g)$ first over elements and then over matrix conjugacy classes. The lift has 32 classes; the adjoint image has five.

representation	τ	$\dim W_\tau$	rank	elements	classes
lift on $\mathbb{C}^2$	(1)	2	0	0	0
	(4)	5	0	0	0
	(8)	9	0	0	0
	$(4,4)$	25	1	1	1
adjoint on $\mathrm{End}(\mathbb{C}^2)$	(1)	4	1	1	1
	(2)	10	2	2	2
	$(1,1)$	16	2	2	2
	(3)	20	2	2	2

All braid, unitarity, closure, integrality, and equality assertions pass. The largest projector error is below 4×10^{-15}.

5 Reproduction and interpretation

Run

```
.venv/bin/python for_this_guy/finite_braid_images/
  ising_clifford/verification.py
.venv/bin/marimo edit for_this_guy/finite_braid_images/
  ising_clifford/notebook.py
```

after joining wrapped lines. The computation verifies the stated formulas for the displayed B_3 Ising representation. It does not claim that every root-of-unity Jones representation has finite image; finiteness remains an essential hypothesis.